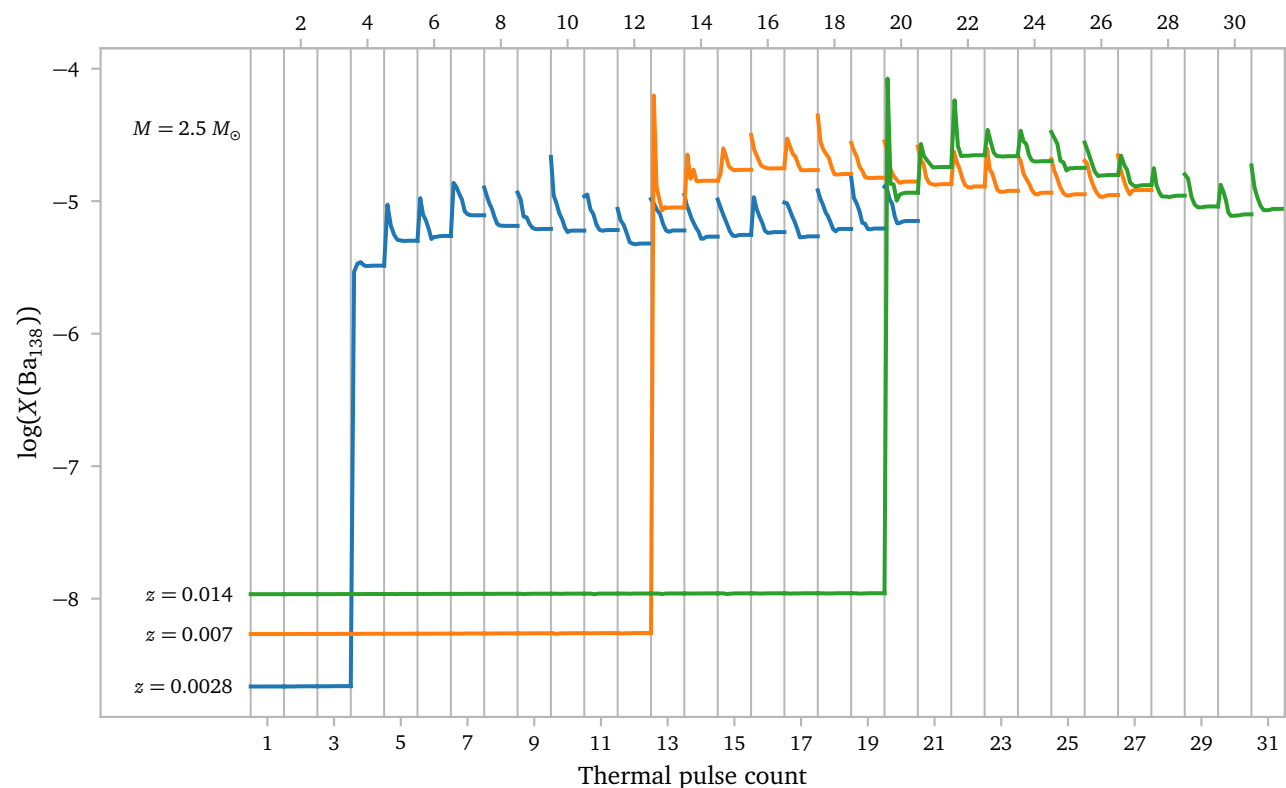


Notes Week 26

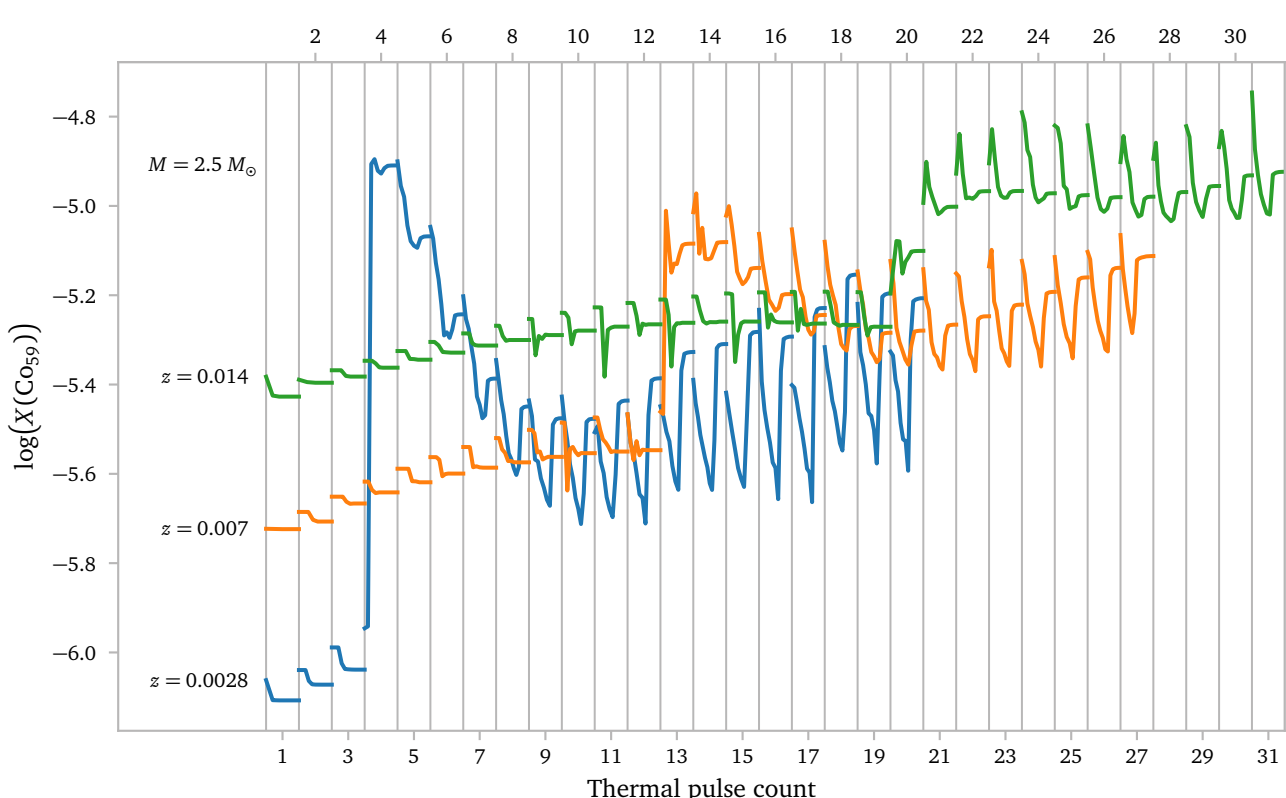
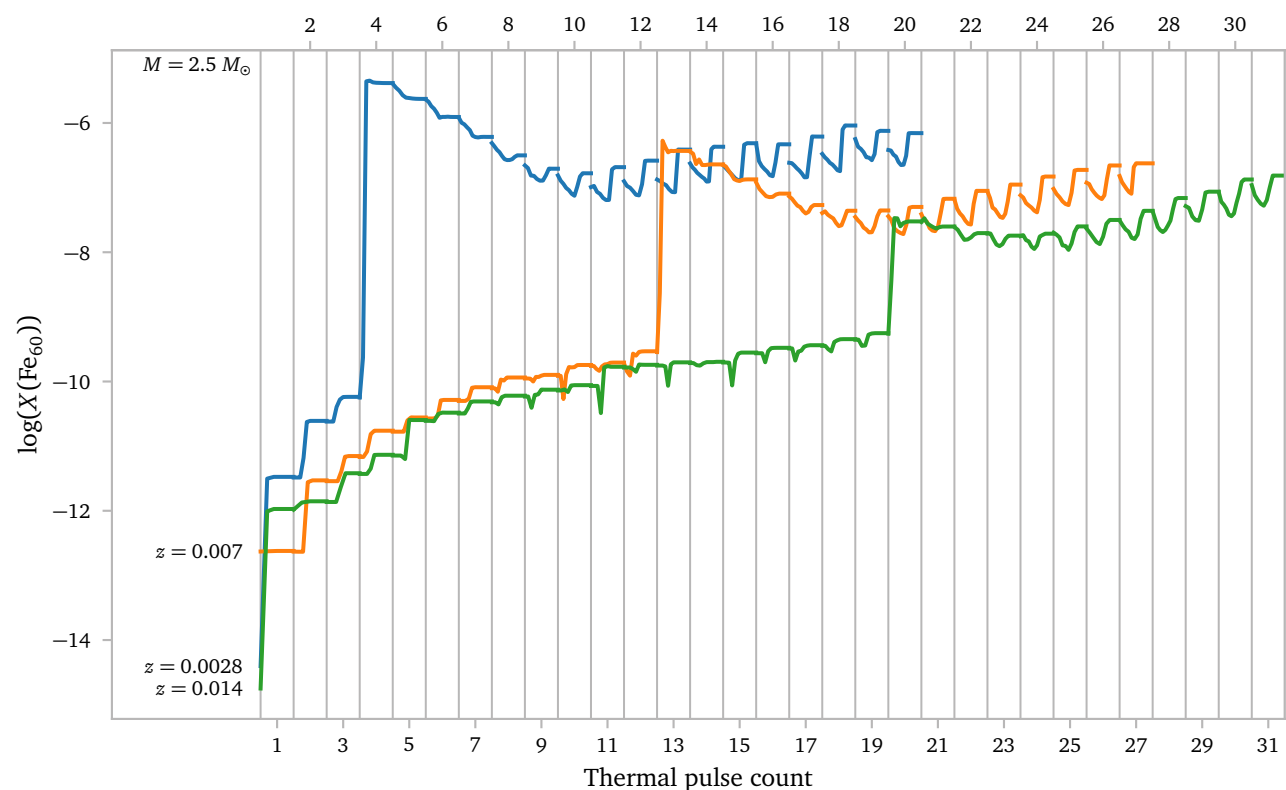
a. Intershell

The plot below shows the effects of metallicity on the the Ba_{138} abundances in the intershell for a $1.5 M_{\odot}$ TPAGB star.

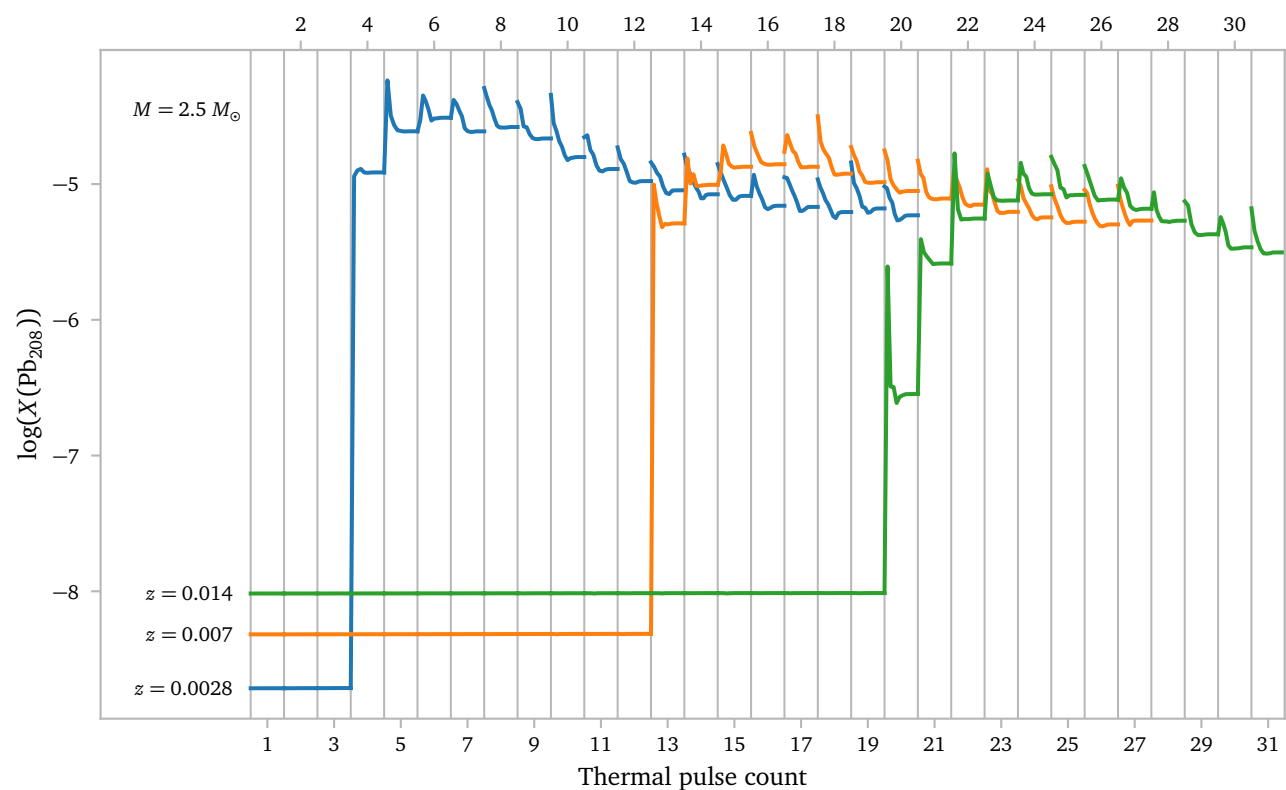


It looks like the metallicity changes the first time dredge-up occurs. This is expected as well, since we see that lower metallicity stars have more dredge-up than the higher metallicity ones (Karakas & Lugaro, 2016).

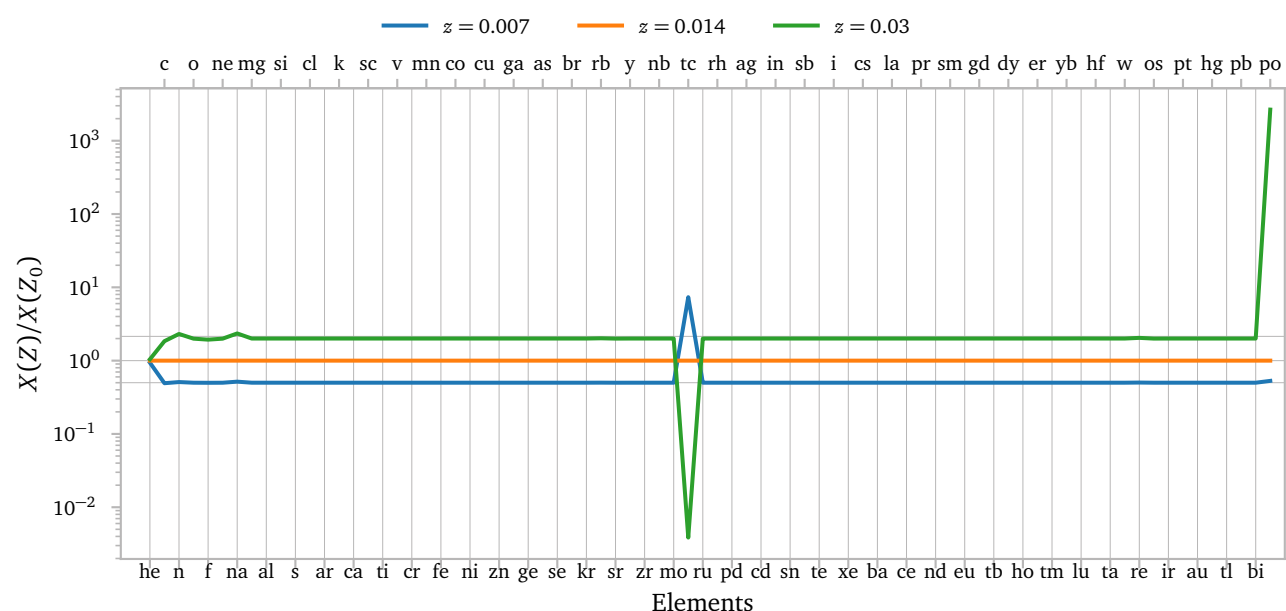
Now let me look at two elements that are very quickly produced by the s-process.



Now I plot Pb_{206} , the heaviest stable isotope that is produced by the s process

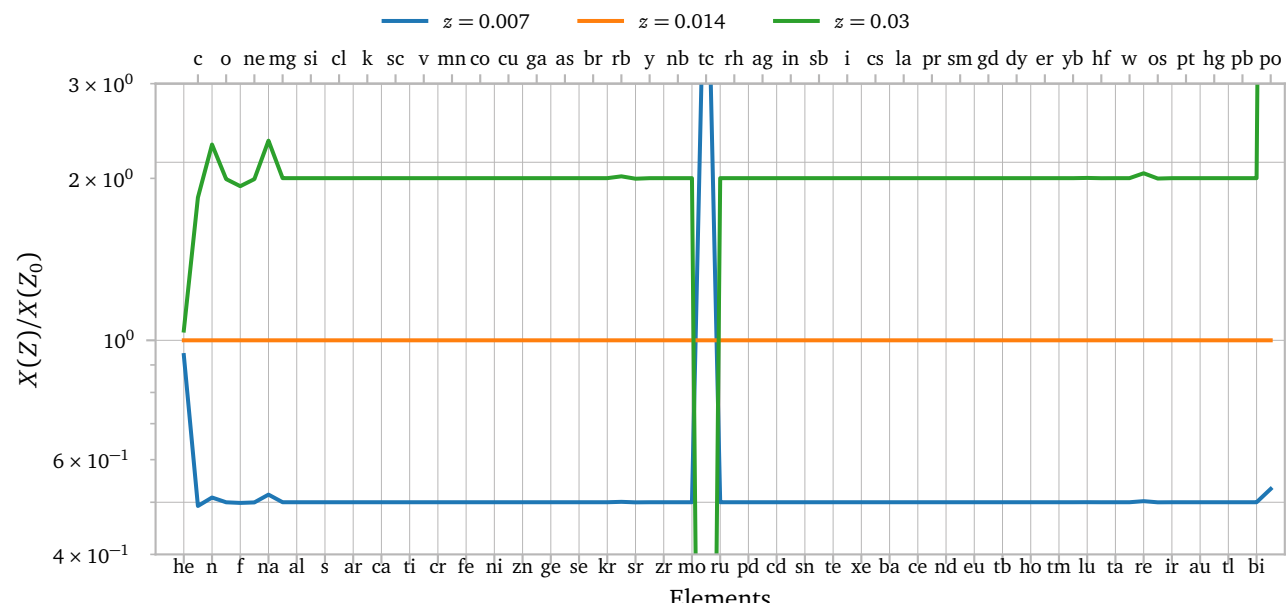


Here I plot the metallicity dependence on the initial envelope abundances of a $1.5 M_{\odot}$ star. This means there is no TDU enrichment. The vertical axis shows the abundances scaled to the solar metallicity model.



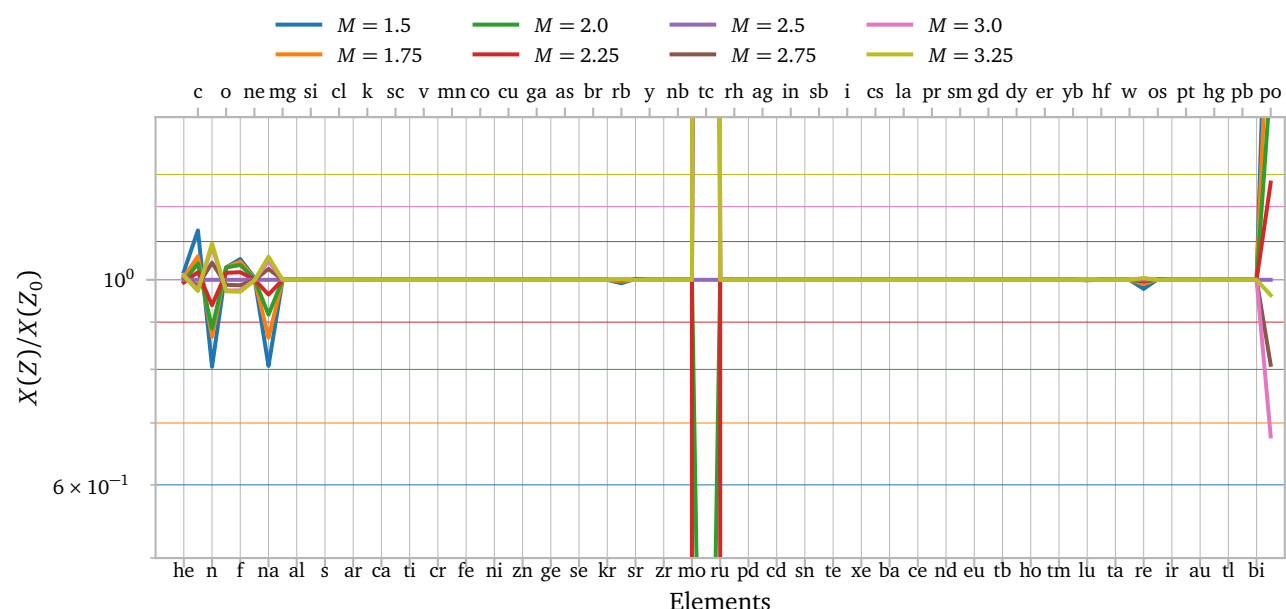
The horizontal lines in this figure show $Z/Z_{\odot}$. Clearly, nearly all elements are just linearly scaled with metallicity. We do see however that the Helium abundances, technetium abundances and the polonium abundances differ. I think this is not a big problem. It just means we should be careful when we analyse the enrichment of these elements.

Below, I show the same figure, now with a zoomed in vertical axis.



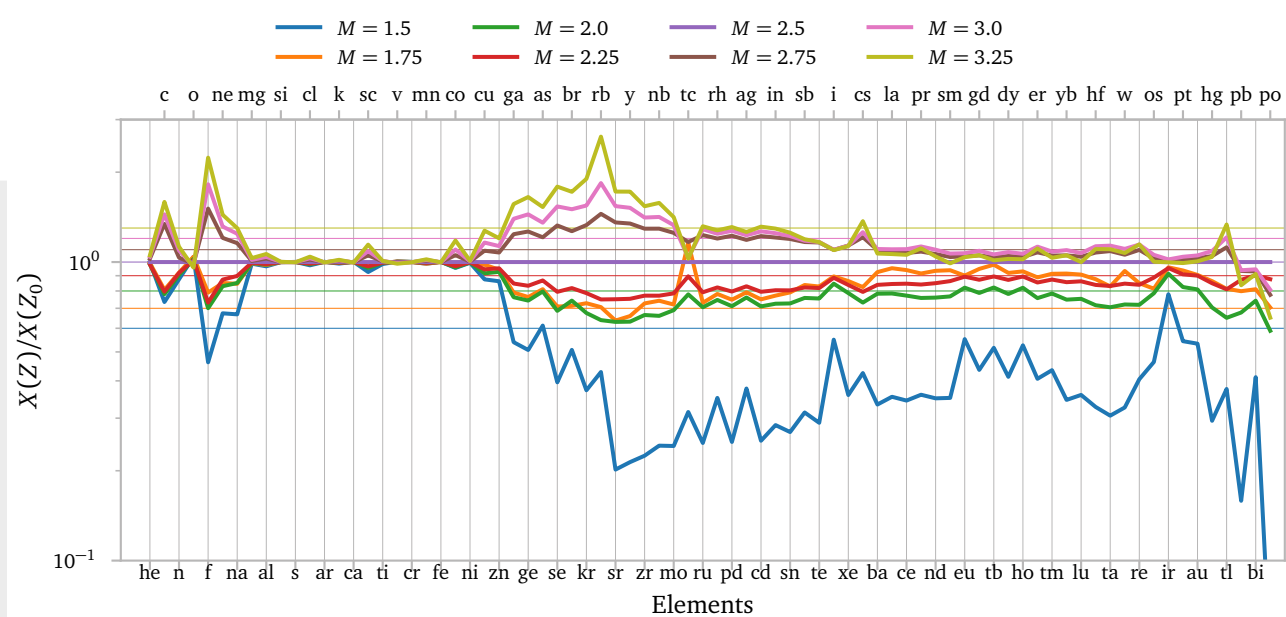
I think we can clearly see here that the $Z = 0.03$ model is scaled by a slightly different value. In fact, it looks like the abundances are nearly all at exactly 2, which would make sense and mean that $Z = 0.028$ or $2Z_{\odot}$.

We can do the same, but now looking at the mass:



We see that for the light elements, there is some variation with mass, but, nearly all s-process elements have initial abundances that are independent of the initial mass of the star.

If we oppose this to the final abundances, we see a clear mass dependence, although not linear and not equal per element.



Anyway, we can simply use the metallicity and linearly scale this with a standard envelope abundance to get the envelope abundances for our MESA models.

References

Karakas, A. I. & Lugaro, M. (2016) *Stellar Yields from Metal-rich Asymptotic Giant Branch Models*. *ApJ*825, 26.